



School of Future Tech

Case Study Report

on

Online Result Processing System

by

Prince Rajesh Yadav

150096725032

Index

1. Introduction to the Case Study.
2. Problem Statement / Case Background (Abstract).
3. Problem Statement / Case Study Design.
4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study.
5. Problem Statement / Case Study Implementation Details and Snapshots.
6. Problem Statement / Case Study Results and Conclusion.
7. References

1. Introduction to the Case Study

Educational institutions always assess their students based on their performance in different subjects. It is very difficult to manage and calculate student results manually.. There is a need to develop a program to process student results efficiently without any human error.

The case study is based on developing an Online Result Processing System using the C++ programming language. The program will take student marks in different subjects, perform arithmetic operations, and calculate the result. It will also check the result status (Pass/Fail).

The project is based on using program logic to perform arithmetic operations and decision-making, which can be used to process student results. The program can be used to process student results using simple programming concepts.

2. Problem Statement / Case Background (Abstract)

Background

In conventional systems, the results of the students are calculated manually by the teacher or administrator. In this process, marks are added, and percentages are calculated, and whether the student has passed or failed is determined individually for each student. This process is time-consuming and involves more chances of error.

With the help of programming, this process can be automated to make the process faster, more accurate, and error-free.

Abstract

The current case study deals with the implementation of an Online Result Processing System using the C++ programming language. In this project, marks are taken for different subjects, arithmetic operations are carried out to calculate the total marks and percentage, and whether the student has passed or failed is determined using the programming process.

In this project, basic programming operations are used, which are as follows:

- Input and Output Operations
- Arithmetic Operations
- Conditional Statements
- Loop Control Structures

The final output will be in the form of a marksheet of the student.

3. Case Study Design

The Online Result Processing System has a well-structured design with the following steps:

****Stage 1: Data Input****

The system prompts the user to enter the marks for five subjects:

1. Mathematics
2. Physics
3. Chemistry
4. English
5. Computer

The marks for each subject will be out of 100.

****Stage 2: Arithmetic Processing****

Next, the system performs arithmetic operations to calculate the following:

- Total Marks
- Percentage

The formulas used for the arithmetic operations are:

- Total Marks = Sum of all the marks of the subjects
- Percentage = Total Marks / Number of Subjects*100

****Stage 3: Result Evaluation****

The system will now calculate the result status of the student based on the percentage.

- If percentage $\geq 40\%$ → PASS
- If percentage $< 40\%$ → FAIL

****Stage 4: Grade Calculation****

The system will calculate the grade based on the percentage.

Percentage	grade
90 and above	A
75-89	B
60-74	C
50-59	D
40-49	E
Below 40	F

****Stage 5: Result Display****

Finally, the system will display the marksheet with the following details:

- Subject marks
- Total marks
- Percentage
- Grade
- Result (Pass/Fail)

4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study

Key Methods and Programming Techniques Used

1. **Input and Output Operations**

The program uses C++ input/output operations as follows:

- cin → to read marks entered by the user
- cout → to display the results and marksheet

2. **Arithmetic Operations**

Arithmetic operations are used to compute the total and percentage.

Arithmetic operators used:

- + → to add marks
- / → to compute the percentage

3. ****Conditional Statements****

Conditional statements are used to compute the grade and result.

Conditional statements used:

- if
- else if
- else

4. ****Loop Structure****

The do...while loop structure is used to develop a menu-driven program that can generate the results repeatedly until the user decides to exit the program.

****Technology Stack Used for the Case Solution****

1. ****Programming Language Used****

- The programming language used to develop the program is C++.

2. ****Library Used****

- The library used to develop the program is iostream.

3. ****Development Environment Used****

- The development environment used to develop the program is VS Code.

5. Problem Statement / Case Study Implementation Details and Snapshots.

The system is implemented in a single C++ program file.

Example:

online_result.cpp

Functions of the program:

- Take marks input
- Perform calculations
- Determine grade and result
- Display marksheet

Snapshots:

1. This is the snapshot after running the code

```
===== ONLINE RESULT PROCESSING SYSTEM =====  
1. Generate Result  
2. Exit  
Enter Choice: █
```

2. This is the whole output after entering marks and it also shows the marsheet.

```
===== ONLINE RESULT PROCESSING SYSTEM =====
1. Generate Result
2. Exit
Enter Choice: 1

Enter Marks (Out of 100)
Maths: 77
Physics: 90
Chemistry: 80
English: 85
Computer: 96

===== MARKSHEET =====
Maths      : 77
Physics    : 90
Chemistry  : 80
English    : 85
Computer   : 96
-----
Total Marks : 428 / 500
Percentage  : 85.6%
Grade       : B
Result      : PASS
=====
```

6. Problem Statement / Case Study Results and Conclusion.

Findings

The Online Result Processing System successfully demonstrates the effectiveness of programming in the calculation of academic student results.

Observations:

- Arithmetic operations successfully calculate the total and percentage.
- Conditional statements successfully determine the pass or fail status.
- Instant and reliable results are generated.
- The program successfully eliminates calculation errors.

Conclusion

This case study demonstrates a basic yet effective system that can be used to process student results using the C++ programming language. The program successfully

incorporates basic programming concepts such as arithmetic, conditional, and loop statements to create a structured marksheet.

The system demonstrates the effectiveness of programming in the academic environment, which can be further enhanced to create a complete student management system that can handle the needs of all students.

7. References

- C++ Programming Language Documentation
- Standard C++ Input Output Library (iostream)
- Basic Programming Concepts (Arithmetic and Conditional Logic)
- Educational resources on student result processing systems